

Mock Heat Flow Study

Author	Example Research Team
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Input	examples/dip/CreateReport/DIPfile

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1 Introduction

1.1 Purpose

This document describes a mock heat-flow experiment. The parameter tables show the **evaluated** values used for one run. All values, people, and publication references are illustrative.

1.2 How to read the report

The flow speed is changed by `overrides.dip`; its effective value and replacement source appear together in the parameter block. The sensor and analysis settings come from reusable schemas. The source and custom-unit sections explain where the other inputs originated.

1.3 Scope

The example includes two schemas, two custom units, multiple source files, publication details, and values with physical units. The linked contents page leads to each main report section.

2 Parameters

experiment.analysis.method

Value	"mean"
Description	Method used to summarize the mock sensor readings
Applied schemas	analysis
Contributing schema	analysis
Overridden	no
Declared at	analysis.dipl:6
Declaration	method str = "mean"

experiment.analysis.window

Value	8
Description	Number of readings in a summary window
Applied schemas	analysis
Contributing schema	analysis
Overridden	no
Declared at	analysis.dipl:8
Declaration	window int = 8

experiment.duration

Value	12
Units	h
Description	Length of one run
Overridden	no
Declared at	parameters.dip:8
Declaration	duration float = 12 h

experiment.flow_speed

Value	3
Units	m*s-1
Description	Nominal inlet speed, replaced for this run
Overridden	yes
Declared at	parameters.dip:16
Declaration	flow_speed float = 2 m/s
Override at	overrides.dip:1
Override	experiment.flow_speed = 3 m/s

experiment.inlet_temperature

Value	295
Units	K
Description	Initial inlet temperature
Overridden	no
Declared at	parameters.dip:18
Declaration	inlet_temperature float = 295 K
experiment.interval	
Value	1
Units	sample_period
Description	Time between sensor samples in the project's custom unit
Overridden	no
Declared at	parameters.dip:12
Declaration	interval float = 1 sample_period
experiment.outlet_temperature	
Value	301
Units	K
Description	Initial outlet temperature
Overridden	no
Declared at	parameters.dip:20
Declaration	outlet_temperature float = 301 K
experiment.sample_count	
Value	48
Description	Number of readings in the sample set
Overridden	no
Declared at	parameters.dip:10
Declaration	sample_count int = 48
experiment.sensor.accuracy	
Value	0.2
Units	K
Description	Stated temperature accuracy
Applied schemas	sensor
Contributing schema	sensor
Overridden	no
Declared at	sensor.dipl:9
Declaration	accuracy float = 0.2 K
experiment.sensor.model	
Value	"T100"
Description	Sensor model identifier
Applied schemas	sensor
Contributing schema	sensor
Overridden	no
Declared at	sensor.dipl:7
Declaration	model str = "T100"
experiment.title	
Value	"Heat & Flow_demo"
Description	A small mock experiment for the generated report.
Overridden	no
Declared at	parameters.dip:2
Declaration	title str = "Heat & Flow_demo"
Authors	A. Example and B. Researcher
Publication	A demonstration of evaluated DIP reports
Year	2026
DOI	10.1234/example.heat-flow

experiment.track_length	
Value	5
Units	lab_length
Description	Length expressed in the custom project unit
Overridden	no
Declared at	parameters.dip:14
Declaration	track_length float = 5 lab_length
lab.operator	
Value	"Example Research Team"
Description	Team responsible for this mock run
Overridden	no
Declared at	environment.dip:6
Declaration	operator str = "Example Research Team"
lab.pressure	
Value	1.013e5
Units	Pa
Description	Ambient pressure during the demonstration
Overridden	no
Declared at	environment.dip:4
Declaration	pressure float = 101325 Pa
lab.room_temperature	
Value	293
Units	K
Description	Ambient laboratory temperature
Overridden	no
Declared at	environment.dip:2
Declaration	room_temperature float = 293 K

3 Schemas

analysis	
Description	Reusable analysis and reporting choices
Declared at	analysis.dipl:6
Authors	C. Analyst
Publication	Illustrative data reduction method
Year	2026
sensor	
Description	Reusable temperature sensor settings
Declared at	sensor.dipl:7
Authors	A. Example and B. Researcher
Publication	A mock sensor calibration study
Year	2026
DOI	10.1234/example.sensor

4 Sources

DIP0	
Content hash	SHA-256: e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
DIP0_FILE0	
Path	sensor.dipl
Parent	DIP0_project0
Parent line	9

Content hash	SHA-256: 3382c36674d1c21311f4f13ed5d7926fcbef0eb726070f29aab96c3f0b15dc68
DIP0_FILE1	
Path	analysis.dipl
Parent	DIP0_project0
Parent line	9
Content hash	SHA-256: a80119fe87f05dcf95b98477457bd43d533bce473f889b7640bfb4609b7ff3d7
DIP0_FILE2	
Path	parameters.dip
Parent	DIP0_project0
Parent line	12
Content hash	SHA-256: fc231633b35df60d9f5ef14380a795016e552e6bd5cc14f5bdca36865531e80a
DIP0_FILE3	
Path	environment.dip
Parent	DIP0_project0
Parent line	12
Content hash	SHA-256: f9d649122600b69254fee4b5bc41714361fe9322d7d41c7634b8fbf70d17a0b5
DIP0_OVERRIDE0	
Path	overrides.dip
Parent	DIP0_project0
Parent line	15
Content hash	SHA-256: 28aa6bc9d74975406b6f73081f4fbf15b060d02d9c5b800a9ee5520688d30c45
DIP0_project0	
Path	DIPfile
Parent	DIP0
Content hash	SHA-256: af7ff59c1f9fd6432e80b5072e4f19b389728499a519125b271f06dd9df79d42

5 Custom units

lab_length	
Definition	2*m
sample_period	
Definition	0.5*s